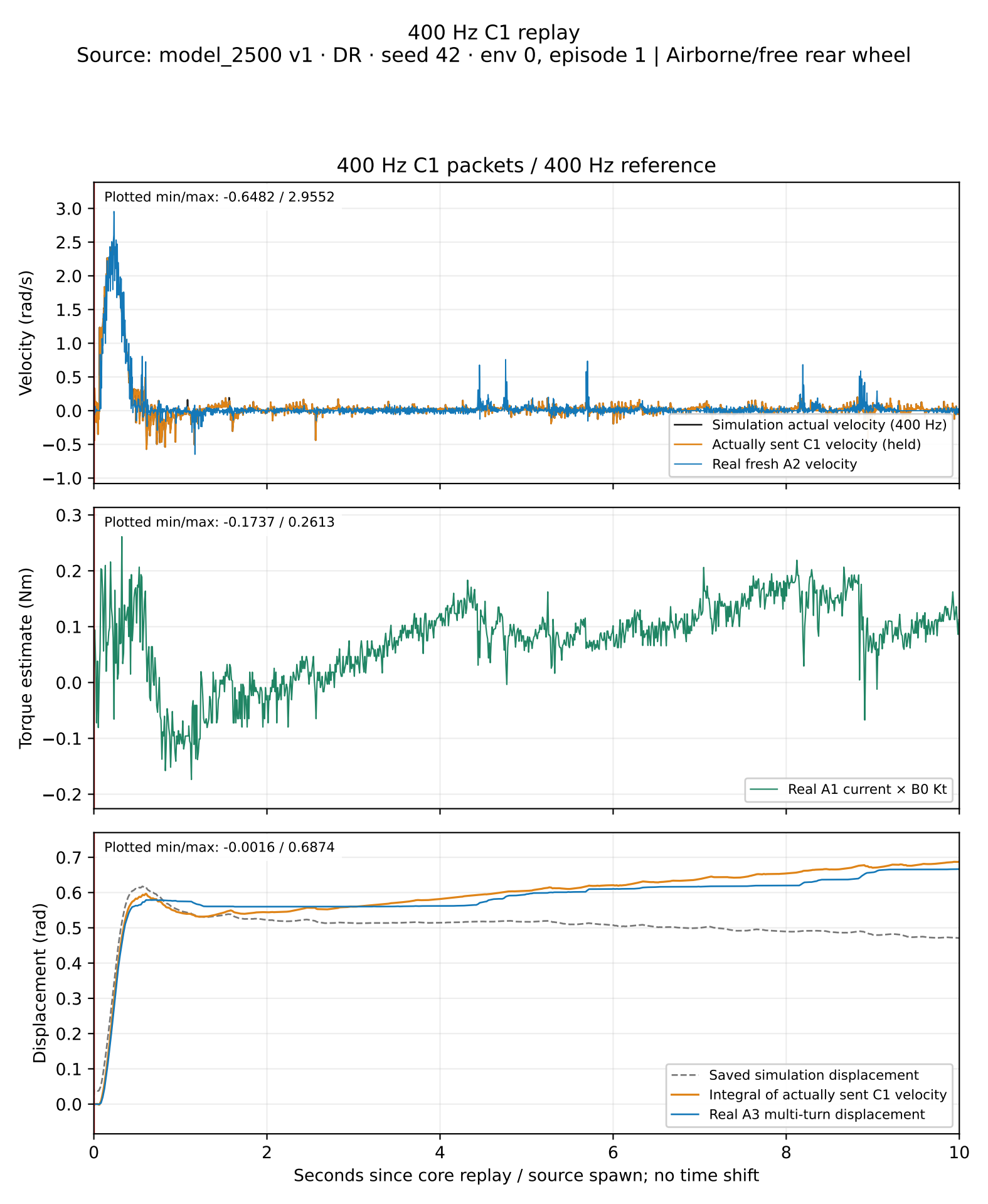
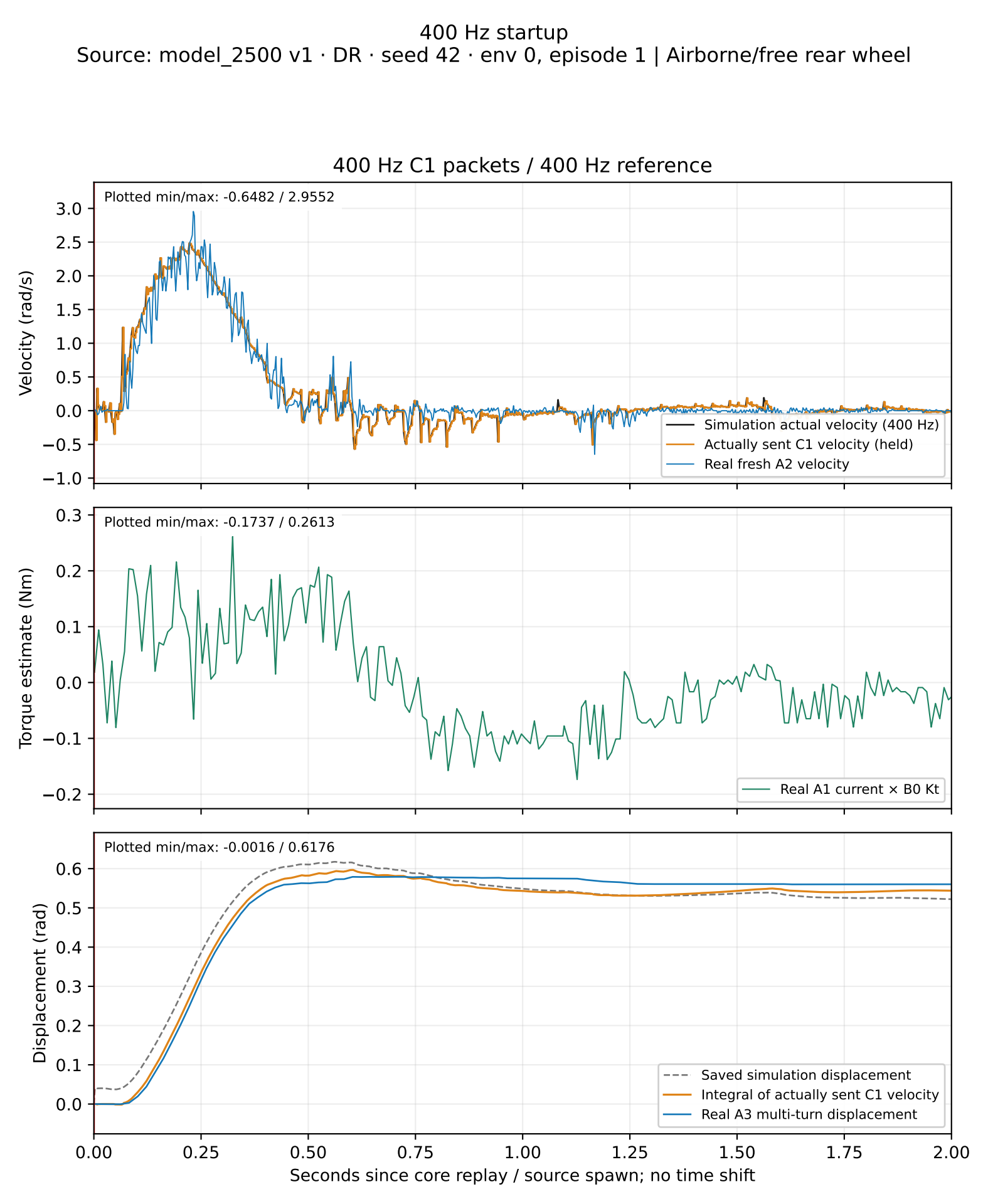
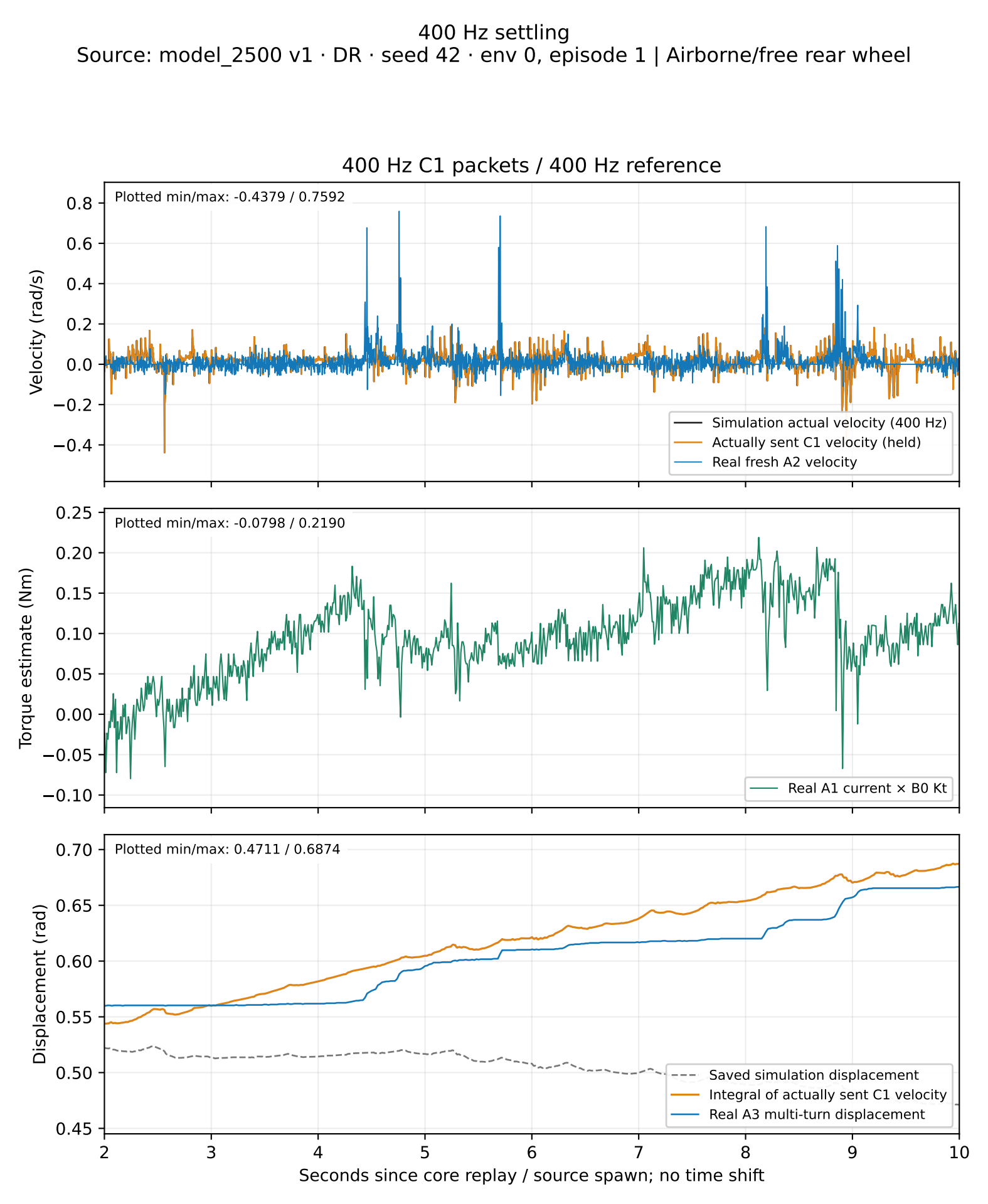
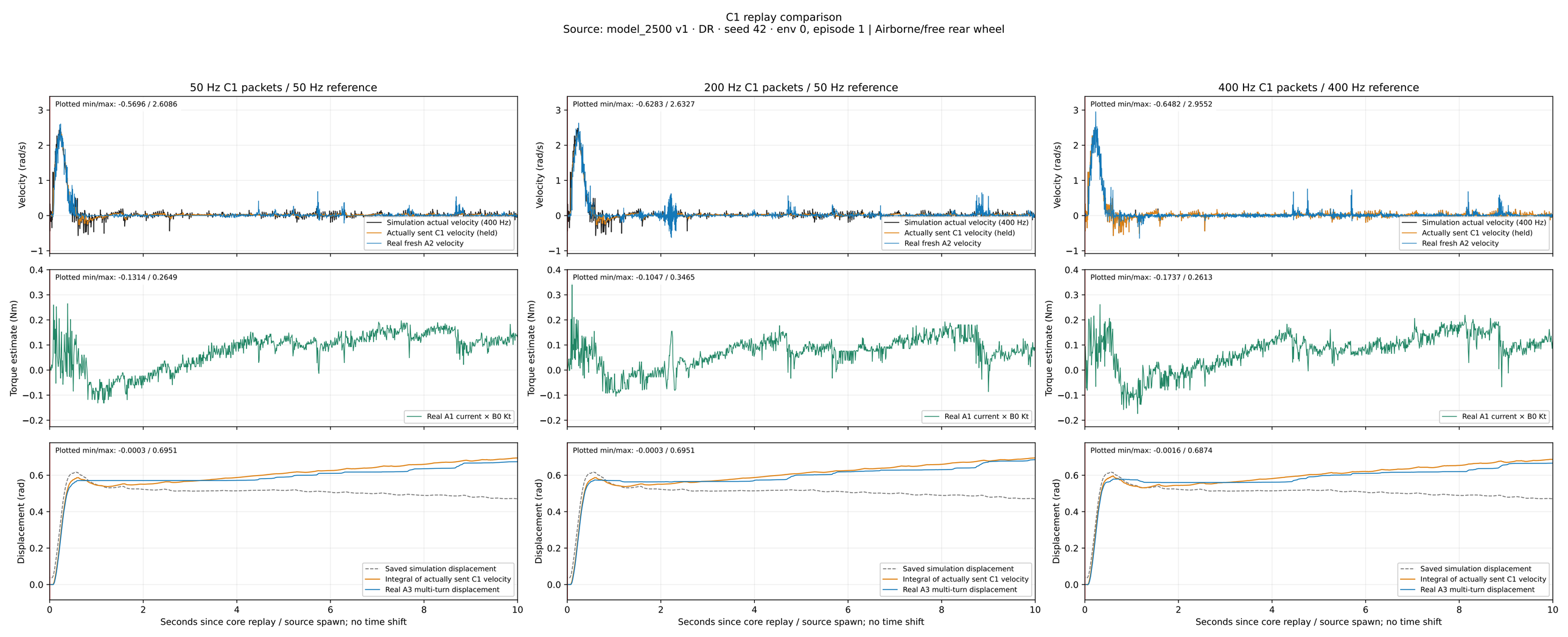
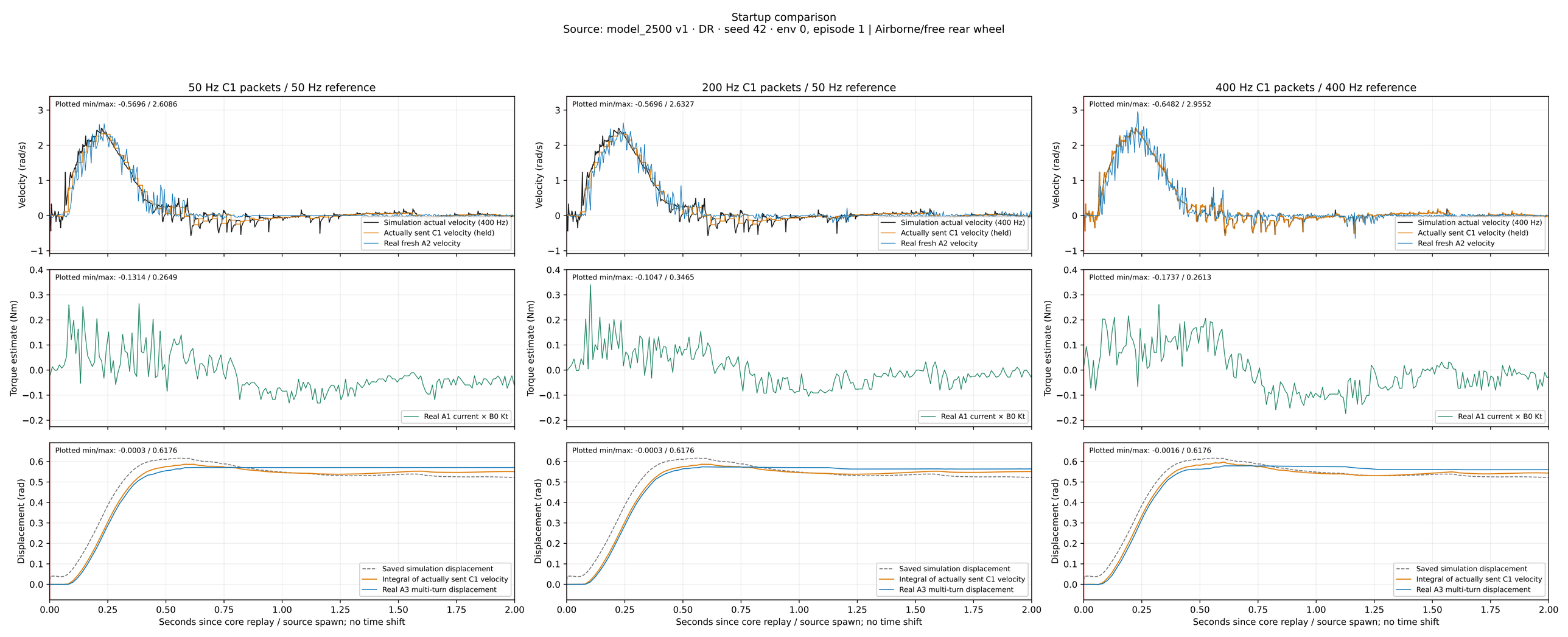
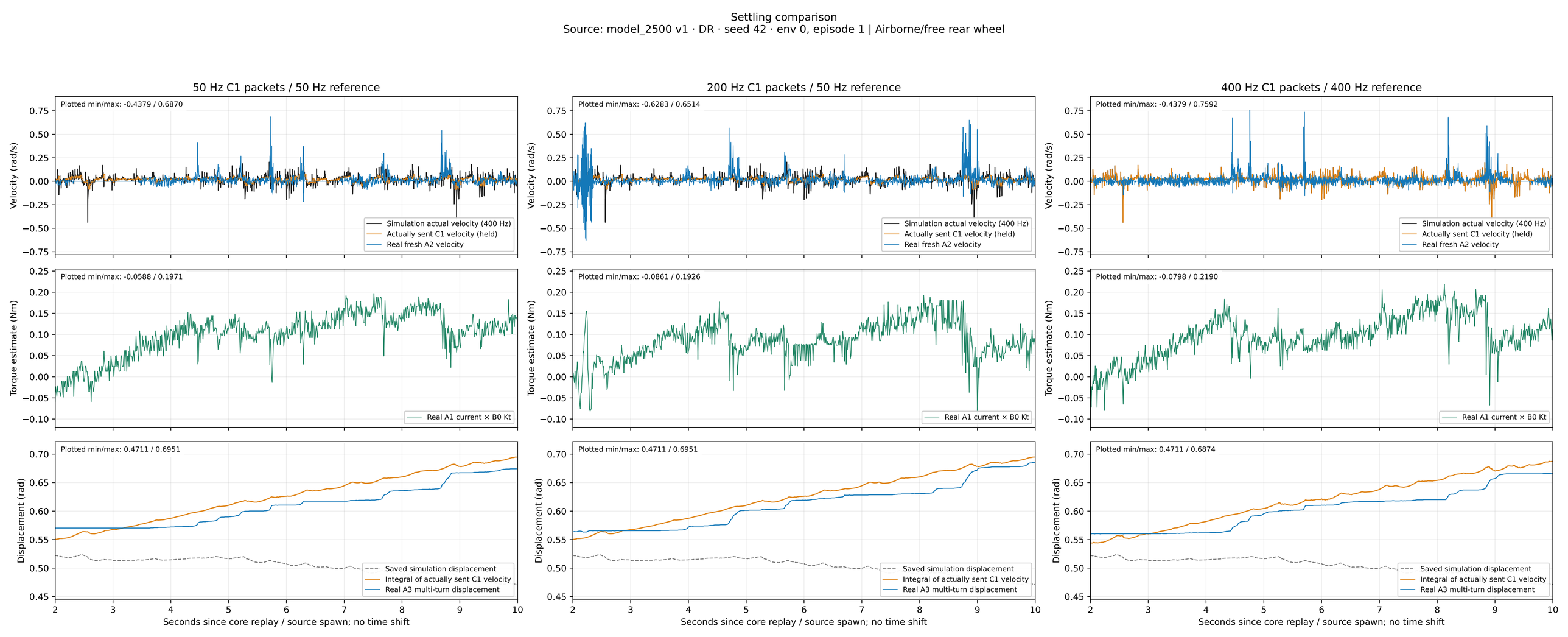

400 Hz C1 replay
Source: model_2500 v1 · DR · seed 42 · env 0, episode 1 | Airborne/free rear wheel
400 Hz C1 packets / 400 Hz reference
Plotted min/max: -0.6482 / 2.9552
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1737 / 0.2613
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0016 / 0.6874
Displacement (rad)
Saved simulation displacement
Integral of actually sent C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift

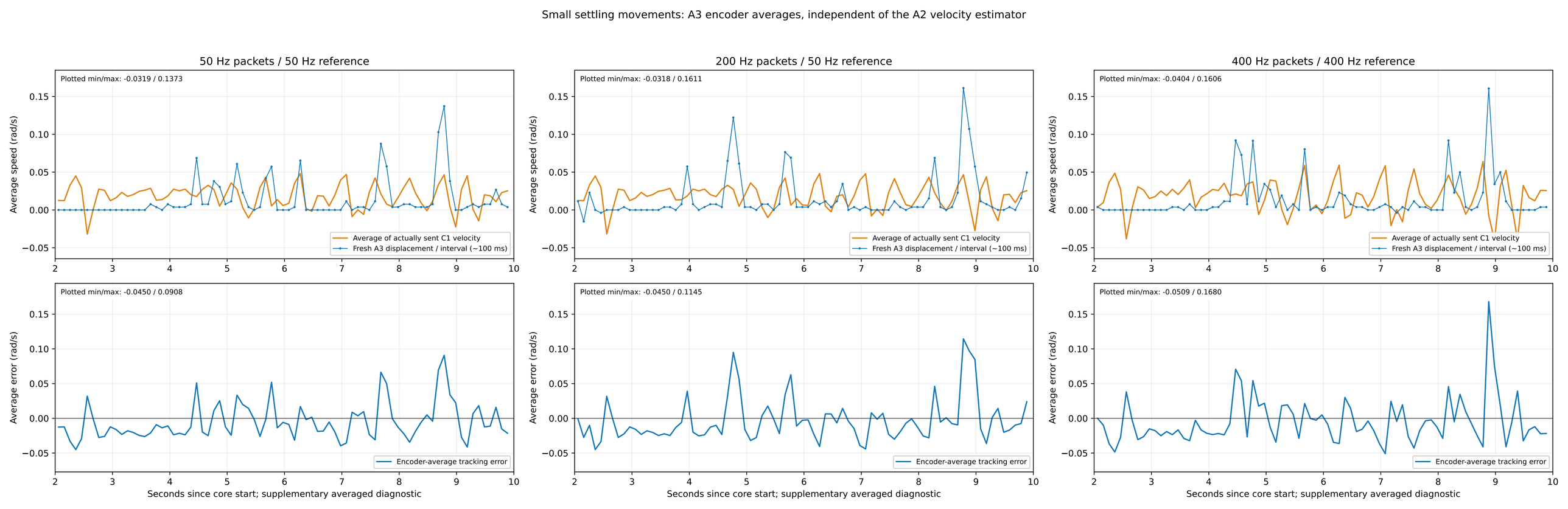

400 Hz startup
Source: model_2500 v1 · DR · seed 42 · env 0, episode 1 | Airborne/free rear wheel
400 Hz C1 packets / 400 Hz reference
Plotted min/max: -0.6482 / 2.9552
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1737 / 0.2613
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0016 / 0.6176
Displacement (rad)
Saved simulation displacement
Integral of actually sent C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift

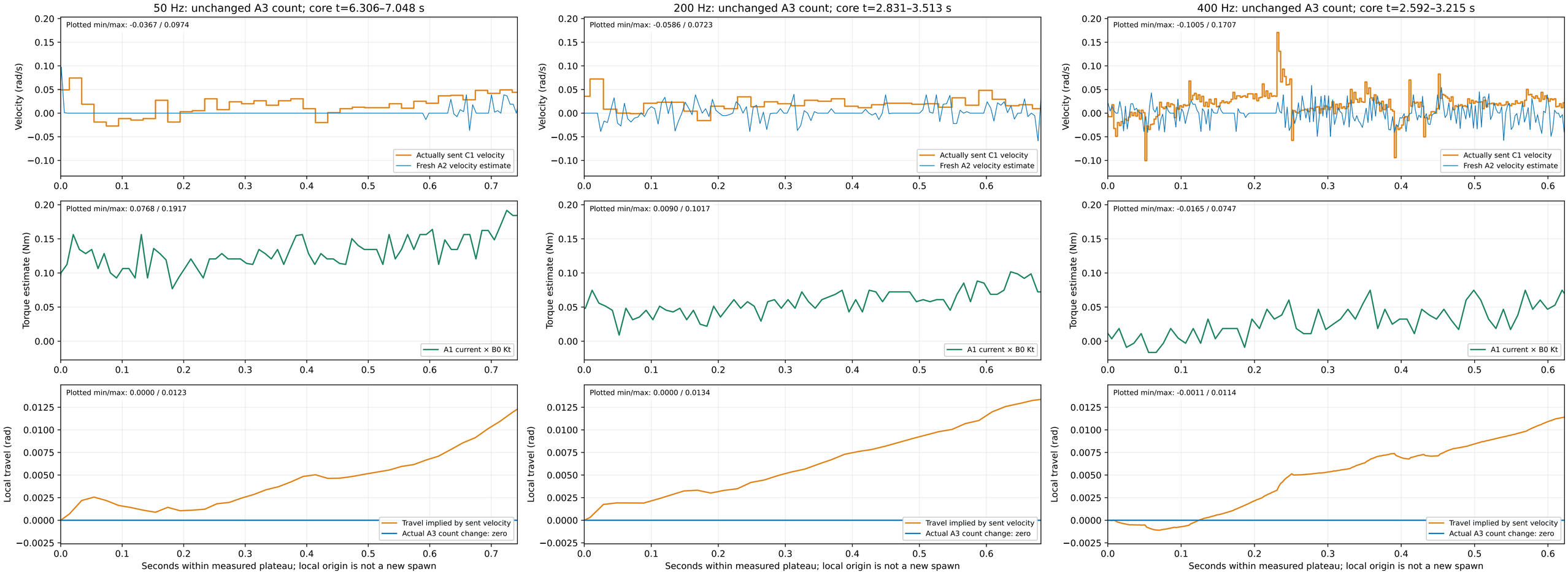

400 Hz settling
Source: model_2500 v1 · DR · seed 42 · env 0, episode 1 | Airborne/free rear wheel
400 Hz C1 packets / 400 Hz reference
Plotted min/max: -0.4379 / 0.7592
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.0798 / 0.2190
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: 0.4711 / 0.6874
Displacement (rad)
Saved simulation displacement
Integral of actually sent C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift

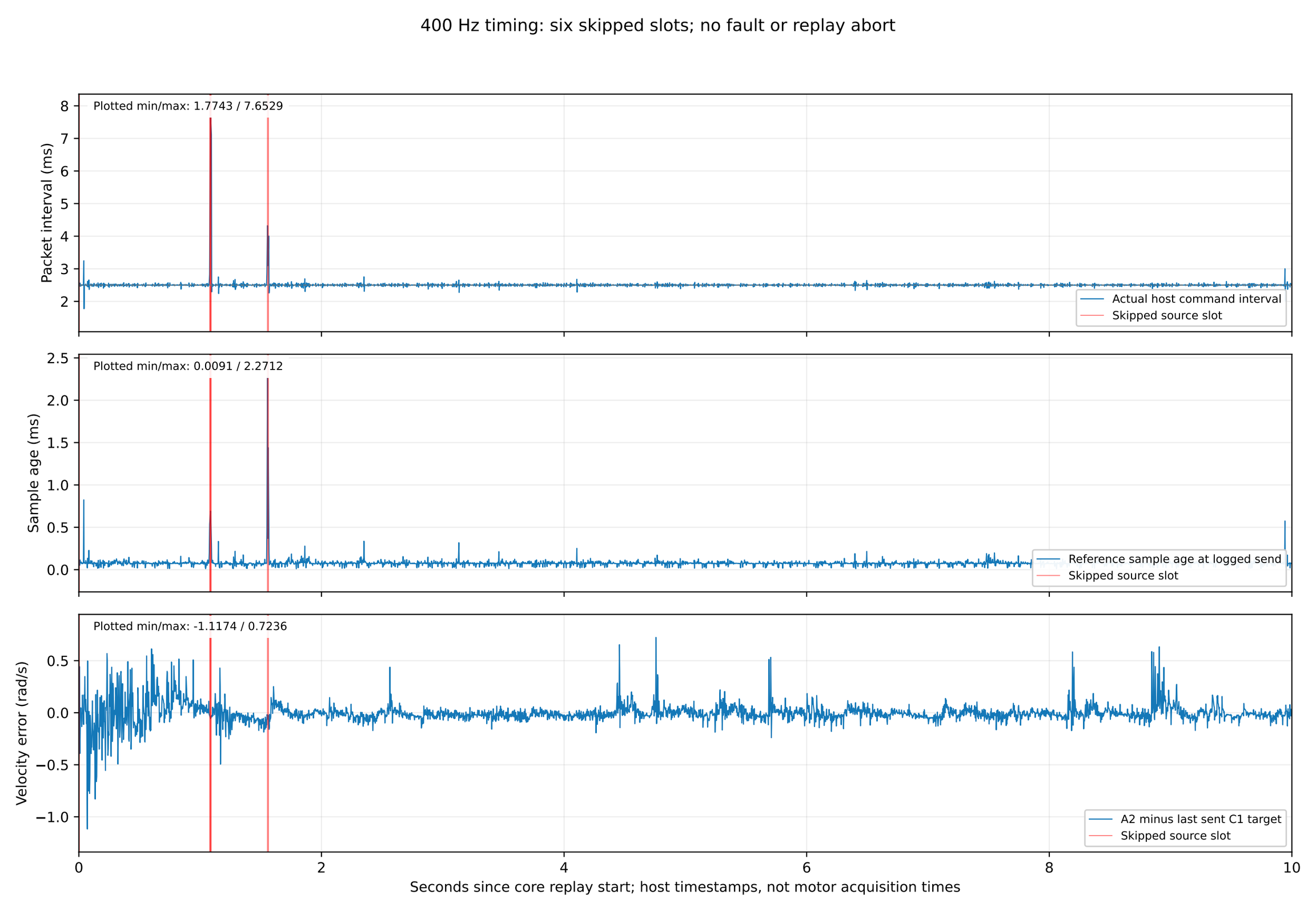

C1 replay comparison
Source: model_2500 v1 · DR · seed 42 · env 0, episode 1 | Airborne/free rear wheel
50 Hz C1 packets / 50 Hz reference
200 Hz C1 packets / 50 Hz reference
400 Hz C1 packets / 400 Hz reference
Plotted min/max: -0.5696 / 2.6086
Plotted min/max: -0.6283 / 2.6327
Plotted min/max: -0.6482 / 2.9552
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1314 / 0.2649
Plotted min/max: -0.1047 / 0.3465
Plotted min/max: -0.1737 / 0.2613
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0003 / 0.6951
Plotted min/max: -0.0003 / 0.6951
Plotted min/max: -0.0016 / 0.6874
Displacement (rad)
Saved simulation displacement
Integral of actually sent C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift


Startup comparison
Source: model_2500 v1 · DR · seed 42 · env 0, episode 1 | Airborne/free rear wheel
50 Hz C1 packets / 50 Hz reference
200 Hz C1 packets / 50 Hz reference
400 Hz C1 packets / 400 Hz reference
Plotted min/max: -0.5696 / 2.6086
Plotted min/max: -0.5696 / 2.6327
Plotted min/max: -0.6482 / 2.9552
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1314 / 0.2649
Plotted min/max: -0.1047 / 0.3465
Plotted min/max: -0.1737 / 0.2613
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0003 / 0.6176
Plotted min/max: -0.0003 / 0.6176
Plotted min/max: -0.0016 / 0.6176
Displacement (rad)
Saved simulation displacement
Integral of actually sent C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift


Settling comparison
Source: model_2500 v1 · DR · seed 42 · env 0, episode 1 | Airborne/free rear wheel
50 Hz C1 packets / 50 Hz reference
200 Hz C1 packets / 50 Hz reference
400 Hz C1 packets / 400 Hz reference
Plotted min/max: -0.4379 / 0.6870
Plotted min/max: -0.6283 / 0.6514
Plotted min/max: -0.4379 / 0.7592
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.0588 / 0.1971
Plotted min/max: -0.0861 / 0.1926
Plotted min/max: -0.0798 / 0.2190
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: 0.4711 / 0.6951
Plotted min/max: 0.4711 / 0.6951
Plotted min/max: 0.4711 / 0.6874
Displacement (rad)
Saved simulation displacement
Integral of actually sent C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift


Small settling movements: A3 encoder averages, independent of the A2 velocity estimator
50 Hz packets / 50 Hz reference
200 Hz packets / 50 Hz reference
400 Hz packets / 400 Hz reference
Plotted min/max: -0.0319 / 0.1373
Plotted min/max: -0.0318 / 0.1611
Plotted min/max: -0.0404 / 0.1606
Average speed (rad/s)
Average of actually sent C1 velocity
Fresh A3 displacement / interval (~100 ms)
Plotted min/max: -0.0450 / 0.0908
Plotted min/max: -0.0450 / 0.1145
Plotted min/max: -0.0509 / 0.1680
Average error (rad/s)
Encoder-average tracking error
Seconds since core start; supplementary averaged diagnostic


Low-speed tracking: longest measured unchanged-encoder interval after t=2 s
50 Hz: unchanged A3 count; core t=6.306–7.048 s
200 Hz: unchanged A3 count; core t=2.831–3.513 s
400 Hz: unchanged A3 count; core t=2.592–3.215 s
Plotted min/max: -0.0367 / 0.0974
Plotted min/max: -0.0586 / 0.0723
Plotted min/max: -0.1005 / 0.1707
Velocity (rad/s)
Actually sent C1 velocity
Fresh A2 velocity estimate
Plotted min/max: 0.0768 / 0.1917
Plotted min/max: 0.0090 / 0.1017
Plotted min/max: -0.0165 / 0.0747
Torque estimate (Nm)
A1 current × B0 Kt
Plotted min/max: 0.0000 / 0.0123
Plotted min/max: 0.0000 / 0.0134
Plotted min/max: -0.0011 / 0.0114
Local travel (rad)
Travel implied by sent velocity
Actual A3 count change: zero
Seconds within measured plateau; local origin is not a new spawn


400 Hz timing: six skipped slots; no fault or replay abort
Plotted min/max: 1.7743 / 7.6529
Packet interval (ms)
Actual host command interval
Skipped source slot
Plotted min/max: 0.0091 / 2.2712
Sample age (ms)
Reference sample age at logged send
Skipped source slot
Plotted min/max: -1.1174 / 0.7236
Velocity error (rad/s)
A2 minus last sent C1 target
Skipped source slot
Seconds since core replay start; host timestamps, not motor acquisition times